

Assignment 2 - Completion Report

Shell Scripting for System Health Monitoring

Objective

- **health_check.sh** created for daily server health checks.
- Script checks: **root disk usage**, **free memory**, and **nginx process** status.
- If disk usage > 80% OR free memory < 100MB OR nginx not running → print **WARNING** to stderr + log to **/var/log/health_warnings.log**.
- Otherwise → print **SUCCESS** messages to stdout.

1. Script Content - health_check.sh

```
[root@ip-172-31-13-169 ec2-user]# cat health_check.sh
#!/bin/bash

LOG_FILE="/var/log/health_warnings.log"

# Create log file if it does not exist
touch "$LOG_FILE"

# Function to log warning to stderr and log file
warning() {
    echo "WARNING: $1" | tee -a "$LOG_FILE" >&2
}

# -----
# Check Disk Usage
# -----
DISK_USAGE=$(df / | awk 'NR==2 {print $5}' | sed 's/%//')

if [ "$DISK_USAGE" -gt 80 ]; then
    warning "Root filesystem disk usage is ${DISK_USAGE}% (above 80%).\"
else
    echo "SUCCESS: Root filesystem disk usage is ${DISK_USAGE}%."
fi

# -----
# Check Free Memory
# -----
FREE_MEMORY=$(free -m | awk '/^Mem:/ {print $7}')

if [ "$FREE_MEMORY" -lt 100 ]; then
    warning "Free memory is ${FREE_MEMORY}MB (below 100MB).\"
else
    echo "SUCCESS: Free memory is ${FREE_MEMORY}MB.\"
fi

# -----
# Check nginx Process
# -----
if pgrep nginx > /dev/null; then
    echo "SUCCESS: nginx process is running.\"
else
    warning "nginx process is not running.\"
fi
```

Figure 1: Content of health_check.sh (cat health_check.sh)

- Sets log file path: **/var/log/health_warnings.log**.
- Defines a **warning()** function → writes to stderr + appends to log file.
- Checks disk usage using **df / | awk**.
- Checks free memory using **free -m | awk**.
- Checks nginx process using **pgrep nginx**.
- Prints SUCCESS or WARNING for each check.

2. Making Script Executable & System Context

```
health_check.sh
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]# ls
health_check.sh
[root@ip-172-31-13-169 ec2-user]# chmod +x health_check.sh
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]# ls -l health_check.sh
-rwxr-xr-x. 1 root root 1070 Aug 27 08:54 health_check.sh
[root@ip-172-31-13-169 ec2-user]# df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p1 30G   1.8G  29G   6% /
[root@ip-172-31-13-169 ec2-user]# free -h
              total        used        free      shared  buff/cache   available
Mem:           3.7Gi       166Mi       3.3Gi       0.0Ki       252Mi       3.4Gi
Swap:           0B           0B           0B
[root@ip-172-31-13-169 ec2-user]# pgrep nginx
[root@ip-172-31-13-169 ec2-user]# sudo systemctl status nginx
Unit nginx.service could not be found.
[root@ip-172-31-13-169 ec2-user]# sudo systemctl status nginx
Unit nginx.service could not be found.
[root@ip-172-31-13-169 ec2-user]# sudo yum install nginx -y
Amazon Linux 2023 Kernel Livepatch repository                               700 kB/s | 69 kB  00:00
Last metadata expiration check: 0:00:01 ago on Thu Aug 27 10:16:02 2026.
Dependencies resolved.
```

Figure 2: `chmod +x health_check.sh`, `ls -l`, `df -h /`, `free -h`, and `nginx` installation

- Made script executable: `chmod +x health_check.sh`.
- Verified permissions with `ls -l` → confirmed executable (`rw-r-xr-x`).
- Checked disk usage: `df -h /` → 6% used (below 80% threshold).
- Checked memory: `free -h` → 3.3GB free (above 100MB threshold).
- `nginx` was not installed → installed using `sudo yum install nginx -y`.

3. Scenario 1 - nginx Stopped (Warning Case)

```
nginx.service - The nginx HTTP and reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-08-27 10:16:23 UTC; 18s ago
Main PID: 56583 (nginx)
Tasks: 3 (limit: 4543)
Memory: 3.4M
CPU: 32ms
CGroup: /system.slice/nginx.service
├─56583 "nginx: master process /usr/sbin/nginx"
├─56584 "nginx: worker process"
└─56585 "nginx: worker process"

Aug 27 10:16:23 ip-172-31-13-169.ap-south-1.compute.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server: (httpd)
Aug 27 10:16:23 ip-172-31-13-169.ap-south-1.compute.internal nginx[56581]: nginx: the configuration file /etc/nginx/nginx.conf is not supported by this version of nginx
Aug 27 10:16:23 ip-172-31-13-169.ap-south-1.compute.internal nginx[56581]: nginx: configuration file /etc/nginx/nginx.conf is not supported by this version of nginx
Aug 27 10:16:23 ip-172-31-13-169.ap-south-1.compute.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server: (httpd)
[root@ip-172-31-13-169 ec2-user]# pgrep nginx
56583
56584
56585
[root@ip-172-31-13-169 ec2-user]# sudo systemctl stop nginx
[root@ip-172-31-13-169 ec2-user]# pgrep nginx
[root@ip-172-31-13-169 ec2-user]# sudo ./health_check.sh
SUCCESS: Root filesystem disk usage is 7%.
SUCCESS: Free memory is 3416MB.
WARNING: nginx process is not running.
[root@ip-172-31-13-169 ec2-user]# cat /var/log/health_warnings.log
WARNING: nginx process is not running.
```

Figure 3: `nginx` running status, `systemctl stop nginx`, script run showing `WARNING`, and log file contents

- Confirmed nginx was running via **systemctl status nginx**.
- Stopped nginx: **sudo systemctl stop nginx**.
- Ran script: **sudo ./health_check.sh**.
- Result: Disk usage → SUCCESS, Free memory → SUCCESS, nginx → **WARNING**.
- Verified log file: **sudo cat /var/log/health_warnings.log** → WARNING entry present.

4. Scenario 2 - nginx Restarted (Success Case)

```

15.252.145.149 (ec2-user)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/home/ec2-user/
Name
.ssh
.bashrc
.bash_profile
.bash_logout
Remote monitoring
Follow terminal folder
ip-172-31-13-169.ap-south-1.compute.internal 0% 0.39 GB / 3.73 GB 0.01 Mb/s 0.00 Mb/s 104 min ec2-user (x2)
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
[WARNING: nginx process is not running.]
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]# sudo systemctl start nginx
[root@ip-172-31-13-169 ec2-user]# pgrep nginx
57942
57943
57944
[root@ip-172-31-13-169 ec2-user]# sudo ./health_check.sh
SUCCESS: Root filesystem disk usage is 7%.
SUCCESS: Free memory is 3412MB.
SUCCESS: nginx process is running.
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#
[root@ip-172-31-13-169 ec2-user]#

```

Figure 4: systemctl start nginx, pgrep confirmation, and successful health_check.sh run

- Started nginx again: **sudo systemctl start nginx**.
- Confirmed process running: **pgrep nginx** → returned PIDs.
- Ran script again: **sudo ./health_check.sh**.
- Result: All three checks → **SUCCESS** (disk, memory, nginx).

5. Summary of Verification

Check	Threshold	Observed Value	Result
Disk usage (/)	> 80% = warning	6% - 7%	SUCCESS
Free memory	< 100MB = warning	~3.3 - 3.4 GB free	SUCCESS
nginx (stopped)	Not running = warning	Stopped	WARNING (logged)
nginx (started)	Running = success	Running	SUCCESS

6. Conclusion

- Script created, made executable, and tested successfully.
- Correctly detects and logs a WARNING when nginx is stopped.
- Correctly shows SUCCESS for all checks when system is healthy.
- All assignment requirements met.